

SATYAJIT SENAPATI

Lead Azure Data & AI Engineer

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PROFESSIONAL SUMMARY

Senior Azure Data & AI Engineer with 8+ years of experience designing, building, and optimizing scalable cloud data platforms and 2+ years of experience developing enterprise Generative AI and intelligent search solutions. Expertise in Azure **Synapse Analytics**, Azure **Databricks**, Azure **Data Factory**, Azure **Data Lake Gen2**, Apache **Spark (PySpark, Spark SQL)**, **Python, SQL, Terraform, Data Vault 2.0, CI/CD** automation, cloud security, and data architecture across healthcare, retail, e-commerce, and telecom domains.

Experienced in designing and developing AI-powered and search-driven applications using **.NET, ASP.NET Core, C#, Blazor, Entity Framework Core, REST APIs, OpenAI APIs, Elasticsearch, Prompt Engineering, Retrieval-Augmented Generation (RAG)**, semantic search, vector search, AI assistants, and AI agent workflows. Proven track record of delivering secure, scalable, and business-critical data, search, and AI solutions.

CORE SKILLS

Azure & Cloud Services: Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Azure Data Lake Gen2, Azure Functions, Azure Key Vault, Azure Machine Learning, Azure DevOps

Generative AI & Intelligent Applications: OpenAI API, Prompt Engineering, AI Assistants, LLM Integration, Retrieval-Augmented Generation (RAG), Vector Search, Semantic Search, AI Agent Workflows

Big Data & Processing: Apache Spark (PySpark, Spark SQL), Delta Lake, Pandas

Application Development: .NET 10, ASP.NET Core, C#, Blazor, REST APIs, Entity Framework Core

Programming & Scripting: Python, SQL, PL/SQL, Unix Shell Scripting

Data Modeling & Warehousing: Data Vault 2.0, Star & Snowflake Schema, Fact & Dimension Modeling

DevOps & Infrastructure as Code: Git, CI/CD Pipelines, Terraform

Databases & Search Technologies: Oracle, SQL Server, Azure SQL Database, Elasticsearch, Cassandra, MongoDB

BI & Visualization: Power BI, Tableau

Cloud Exposure: AWS (S3, EC2, EMR, Glue, Lambda, Redshift)

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions | Client: United Health Group (Optum)

Lead Azure Data & AI Engineer | Oct 2023 – Present

- Architected cloud-native data, AI, and search solutions leveraging Azure services, OpenAI APIs, Elasticsearch, and .NET technologies.
- Designed and implemented scalable Azure Synapse pipelines ingesting data from on-prem SQL Server, SharePoint Lists, Adobe Analytics, REST APIs, and secure ADLS landing zones.
- Developed reusable BTS and STG pipeline frameworks using parameterized activities and stored procedures, reducing onboarding effort for new data sources.
- Built PySpark and Pandas notebooks for data preprocessing and transformations prior to downstream business layers.
- Designed Gold-layer tables following Data Vault 2.0 standards and developed transformations across bizvault, datamart, and BI layers.
- Integrated ML pipelines into Azure Synapse to enable seamless data flow for advanced analytics use cases.
- Designed and developed AI-powered web applications using .NET, ASP.NET Core, Blazor, and OpenAI APIs to deliver intelligent and conversational user experiences.
- Implemented Retrieval-Augmented Generation (RAG) patterns and AI agent workflows by integrating enterprise data sources with Large Language Models (LLMs).
- Built enterprise search solutions using Elasticsearch and .NET, enabling scalable full-text, semantic, and filtered search capabilities across business datasets.
- Developed Azure Function Apps to externalize data securely through API endpoints.
- Implemented CI/CD pipelines using GitHub Actions to deploy Synapse, Functions, .NET applications, and related artifacts across dev, test, and prod environments.
- Automated Azure infrastructure provisioning using Terraform, including Synapse, ADLS, Key Vault, Azure ML, and Function Apps.
- Implemented enterprise-grade security using Private Endpoints, NSGs, route tables, and secure network configurations.
- Collaborated with business stakeholders, architects, and engineering teams to define technical roadmaps, solution architecture, and delivery strategies.
- Provided technical leadership, code reviews, and mentoring to engineering teams, promoting best practices in data engineering, AI development, and DevOps.

Microland Limited | Client: Mandai (Wildlife)

Azure Data Engineer | Feb 2023 – Sep 2023

- Designed and implemented Databricks workflows for full and incremental ingestion from ADLS into Delta Lake using PySpark and Spark SQL.
- Developed transformation notebooks to build Fact and Dimension tables, orchestrated via modular parent–child notebook design.
- Created configuration-driven frameworks in Databricks to manage source metadata, incremental logic, and notebook execution paths.
- Optimized Databricks workflows using parallel execution and parameterized notebooks to improve performance and scalability.
- Built Azure Data Factory pipelines for full and incremental loads from on-prem SQL Server to ADLS Bronze layer.
- Configured Self-Hosted Integration Runtime (SHIR) and Logic Apps for monitoring and operational automation.
- Implemented CI/CD using GitHub Actions, deploying ADF artifacts and Databricks workflows via Azure CLI and Databricks CLI.

TEKsystems Global Services | Client: Spreetail (E-commerce)

Azure Databricks Engineer | Aug 2021 – Jan 2023

- Designed end-to-end Databricks-based data pipelines ingesting data from Azure SQL, AWS S3, and ADLS into Raw, Bronze, and Silver layers.
- Implemented data cleansing, schema enforcement, and Delta table creation in the Bronze layer.
- Built Fact and Dimension tables in Silver layer to support downstream analytics and reporting.
- Designed and implemented Azure Synapse pipelines, including worker pipelines, batching strategies, and audit frameworks.
- Developed transformation frameworks and reusable notebooks to build Fact and Dimension models, orchestrated through modular parent–child notebook architectures.
- Developed SQL procedures and audit tables in Azure SQL to support pipeline execution and monitoring.
- Deployed Synapse and Databricks artifacts using Azure DevOps CI/CD pipelines across dev and prod environments.
- Collaborated with business stakeholders and solution architects to translate data requirements into scalable cloud-native data solutions and best-practice implementations.

Tata Consultancy Services (TCS) | Client: Ericsson (Telecom)

Azure Databricks Engineer | May 2020 – Jul 2021

- Developed Databricks notebooks using PySpark to load and merge data into Delta tables with full and incremental logic.
- Designed transformation pipelines to consolidate data across financial quarters and business categories.
- Scheduled and orchestrated Databricks jobs for sequential and dependent workloads.

Big Data Developer | Jun 2018 – Apr 2020

- Performed large-scale ETL using Apache Spark and Python on data sourced from AWS S3.
- Built data transformation pipelines and delivered curated datasets for analytical use cases.
- Developed machine learning models for traffic prediction, anomaly detection, and fault management using Spark and Scikit-learn.
- Created analytical dashboards using Tableau and Python visualization libraries.
- Built REST APIs using Flask and Cassandra, enabling data access for UI applications.

Database Administrator | Feb 2017 – May 2018

- Installed, upgraded, and maintained Oracle and SQL Server databases across Azure and on-prem environments.
- Developed PL/SQL procedures and Unix shell scripts for database maintenance and monitoring.

EDUCATION

B. Tech – Electrical Engineering

CV Raman College of Engineering, BPUT | 2016

CGPA: 8.41